

Design/analyze a DP, greedy, approximate algo for Seen/unseen problems

e.g. 1) write DFS algo for an undirected graph, 2) write and analyze the algo for printing path from source to destination using π attributes of nodes, 3) explain coin change problem and write its greedy algo, 4) how to compute the length of LCS, and 5) how to obtain the entries of LCS, 6) solve the following in $\lg n$ time, etc etc

Rod cut, Levenshtein, Dijkstra, Prims, Continuous Knapsack, coin change,

Explain Binary search algorithm (in all aspects covered in this course)

What are the three classes of problems, why satisfiability problem is significant

What are the elements of greedy approach

What are the two ways, the inefficiency added by overlapping subproblems be addressed

What is the approximate solution of the knapsack problem

Execute BFS on the a graph (when needed, scan vertices from top to bottom and edges from left to right)

Computer **keys** and **Ts** for nodes of a graph after running PRIMS algorithm on it

Demonstrate the maximum sub-array sum problem on a suitable problem instance

Edit distance problem (DP), codes generation (greedy)

How single source and all pair shortest path algorithms related to each other

Differentiate d/c and dp, and dp and greedy

Why optimization problems are of special concerns

How the three types of optimization problem w.r.t. discussion in class for DP, differs from each other

What methodology is used to prove the correctness in graph algorithms: loop invariants or something other

What is the issue with negative weight cycles in graphs